
Software Requirements Specification

for

FlowGuard : Business Cashflow Prediction & Monitoring App

Version 1.0 approved

Prepared by Group 16

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Revision History

Name	Date	Reason For Changes	Version
Group 16	25/02/26	Introduction	1.0

1. Introduction

1.1 Purpose

Dokumen Software Requirements Specification (SRS) ini mendeskripsikan kebutuhan perangkat lunak untuk FlowGuard : Business Cashflow Prediction & Monitoring App, Versi 1.0.

FlowGuard adalah aplikasi berbasis web yang dirancang untuk membantu usaha kecil dan menengah (UKM/UMKM) dalam memantau, memprediksi, dan mengelola arus kas bisnis secara lebih terstruktur dan terukur.

Dokumen ini mencakup seluruh ruang lingkup sistem FlowGuard, termasuk:

- Modul Autentikasi dan Otorisasi Pengguna
- Modul Manajemen Pemasukan dan Pengeluaran
- Modul Prediksi Arus Kas
- Modul Simulasi Skenario Keuangan
- Dashboard Laporan dan Analitik
- Sistem Notifikasi dan Peringatan Risiko

SRS ini mendefinisikan kebutuhan fungsional, kebutuhan non-fungsional, batasan sistem, serta karakteristik kualitas yang harus dipenuhi selama proses pengembangan.

1.2 Document Conventions

Dokumen SRS ini menggunakan konvensi sebagai berikut:

- Kata “**harus**” menunjukkan kebutuhan yang bersifat wajib (mandatory).
- Kata “**sebaiknya**” menunjukkan kebutuhan yang direkomendasikan.
- Kata “**dapat**” menunjukkan kebutuhan yang bersifat opsional.

Penomoran kebutuhan menggunakan format berikut:

- Kebutuhan Fungsional: **FR-XX** (contoh: FR-01, FR-02)

- Kebutuhan Non-Fungsional: **NFR-XX**
- Batasan Sistem: **SC-XX**

Setiap kebutuhan memiliki identifikasi unik. Apabila terdapat perubahan prioritas atau revisi kebutuhan, maka perubahan tersebut akan didokumentasikan pada versi dokumen berikutnya.

1.3 Intended Audience and Reading Suggestions

Dokumen SRS ini ditujukan untuk pihak-pihak berikut:

1. Developer

Mengimplementasikan sistem sesuai kebutuhan yang telah ditetapkan.
Disarankan membaca Bagian 2 (Overall Description), Bagian 3 (External Interface Requirements), Bagian 4 (System Features), dan Bagian 5 (Nonfunctional Requirements).

2. System Architect

Merancang arsitektur sistem dan menentukan pendekatan teknis.
Disarankan membaca Bagian 2, Bagian 3, dan Bagian 5.

3. Project Manager

Mengelola ruang lingkup, jadwal, dan sumber daya proyek.
Disarankan membaca Bagian 1 dan Bagian 2.

4. Quality Assurance (QA)

Menyusun dan melaksanakan pengujian sistem.
Disarankan membaca Bagian 4 dan Bagian 5.

5. Stakeholder / Product Owner

Memvalidasi kesesuaian sistem dengan kebutuhan bisnis.
Disarankan membaca Bagian 1, Bagian 2, dan ringkasan Bagian 4.

Dokumen ini disusun secara bertahap dari gambaran umum sistem (Bagian 1–2), kebutuhan antarmuka (Bagian 3), fitur sistem (Bagian 4), hingga kebutuhan non-fungsional dan batasan lainnya (Bagian 5–6).

1.4 Product Scope

Business Cashflow Predictor System adalah aplikasi berbasis web yang dirancang untuk membantu pemilik bisnis dalam mengelola, memantau, dan memprediksi arus kas perusahaan secara sistematis dan proaktif.

Tujuan utama perangkat lunak ini adalah memberikan visibilitas yang jelas terhadap kondisi keuangan bisnis, khususnya terkait ketersediaan kas dalam jangka pendek hingga menengah (30–90 hari ke depan). Sistem ini membantu pengguna mengidentifikasi potensi kekurangan kas

sebelum terjadi, sehingga keputusan keuangan dapat diambil berdasarkan data dan proyeksi yang terukur.

Tujuan dan Sasaran Sistem

Sistem ini dikembangkan untuk mencapai sasaran berikut:

1. Membantu bisnis memantau pemasukan, pengeluaran, piutang, dan hutang secara terstruktur.
2. Menyediakan fitur prediksi arus kas berdasarkan data historis dan transaksi terjadwal.
3. Memberikan peringatan dini apabila diprediksi terjadi defisit kas.
4. Menyediakan fitur simulasi (what-if analysis) untuk mendukung pengambilan keputusan finansial.
5. Mengurangi ketergantungan pada pencatatan manual berbasis spreadsheet yang rawan kesalahan.

Manfaat bagi Bisnis

Implementasi sistem ini memberikan manfaat sebagai berikut:

- Meningkatkan kontrol terhadap arus kas perusahaan.
- Mengurangi risiko gangguan operasional akibat kekurangan dana.
- Membantu perencanaan pembayaran kewajiban dan penagihan piutang secara lebih efektif.
- Mendukung pengambilan keputusan strategis berbasis data keuangan yang terproyeksi.

Keterkaitan dengan Strategi Perusahaan

Sistem ini mendukung strategi bisnis perusahaan dalam hal:

- Penguatan manajemen modal kerja (working capital management).
- Peningkatan efisiensi dan transparansi pengelolaan keuangan.
- Transformasi digital proses administrasi dan pengambilan keputusan finansial.
- Mitigasi risiko operasional akibat ketidakstabilan arus kas.

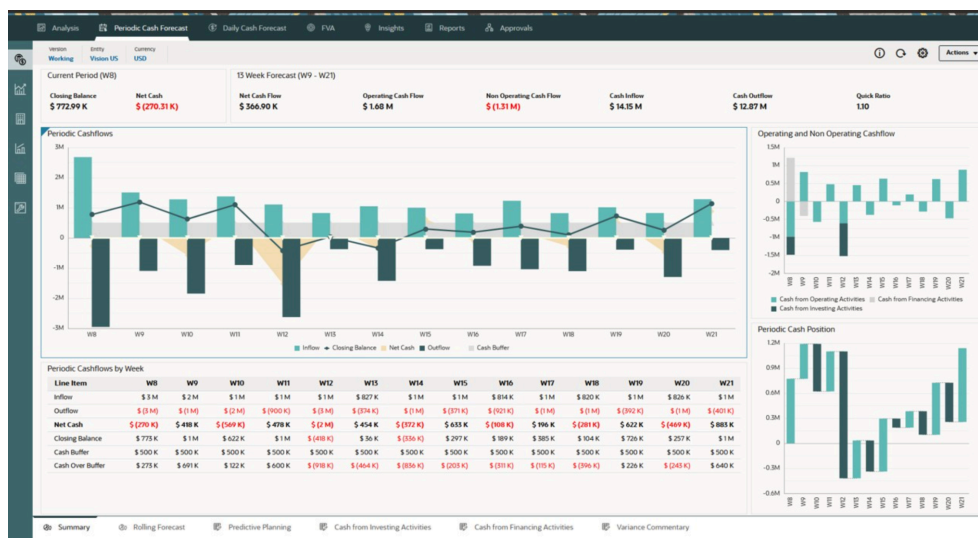
Pada tahap awal (MVP), sistem difokuskan pada fitur inti prediksi arus kas dan pengelolaan transaksi dasar. Pengembangan lanjutan dapat mencakup integrasi dengan sistem akuntansi, integrasi perbankan, serta fitur analitik yang lebih kompleks.

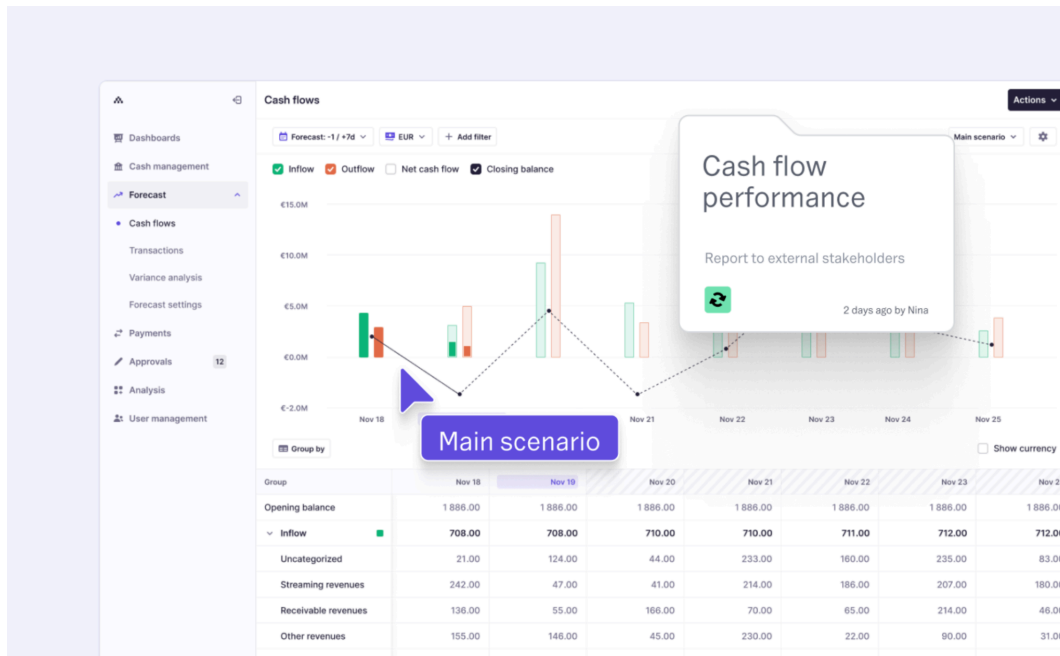
Apabila tersedia, dokumen Vision and Scope yang terpisah dapat dijadikan referensi tambahan untuk detail strategi produk jangka panjang agar tidak terjadi duplikasi informasi dalam dokumen ini.

1.5 References

<https://www.gtreasury.com/solutions/cash/cash-flow-forecasting>

<https://www.atlar.com/guides/the-ultimate-guide-to-the-13-week-cash-flow-forecast>

[illegible]



2. Overall Description

2.1 Product Perspective

FlowGuard is a new, self-contained web-based application specifically designed to meet the financial management needs of small and medium enterprises (SMEs). This application serves as a replacement for error-prone manual spreadsheet tracking. In the MVP (Minimum Viable Product) stage, the system focuses on core features, namely cash flow prediction and basic transaction management, with a long-term plan to integrate with accounting systems and open banking

2.2 Product Functions

- Based on the system's scope, the primary functions users can perform include:
- Secure user authentication and authorization.
- Structured recording and management of cash flows (income, expenses, accounts receivable, and accounts payable).
- Viewing cash flow predictions for the short to medium term (30–90 days ahead) based on historical data and scheduled transactions.
- Running financial scenario simulations (what-if analysis) to support strategic decisions.
- Monitoring data visualizations through a reporting and analytics dashboard.
- Receiving early notifications or risk alerts if the system detects a potential future cash deficit

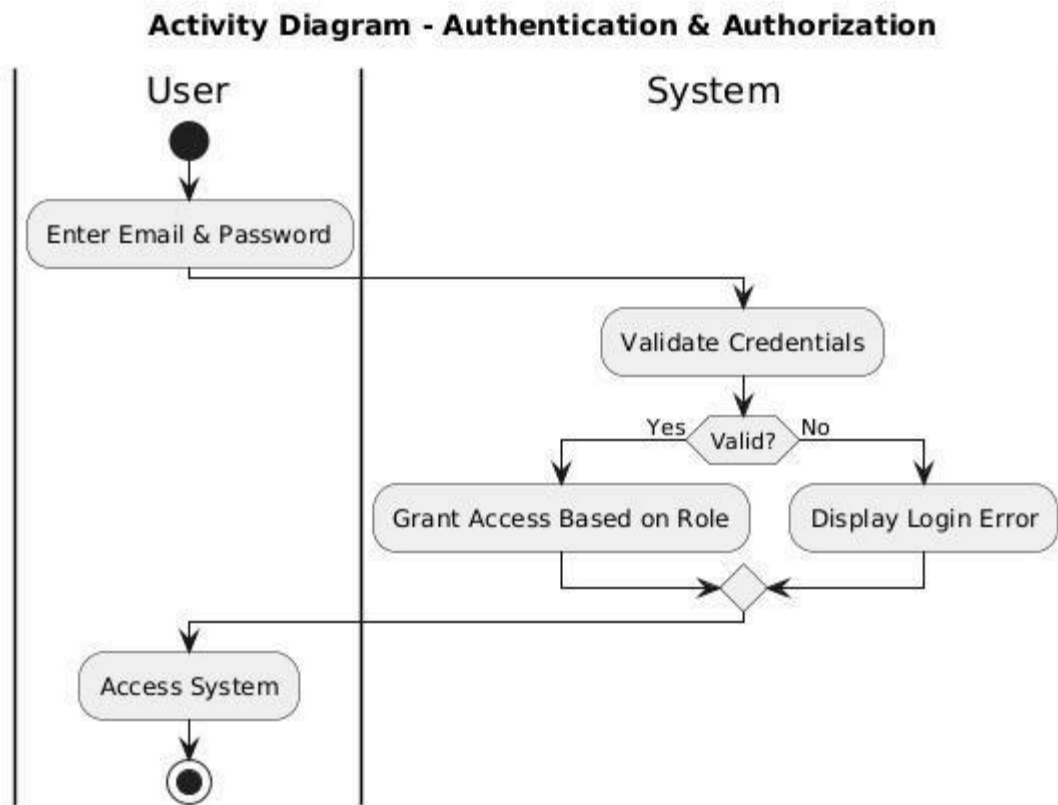


Figure 2.1 Activity Diagram - Authentication & Authorization

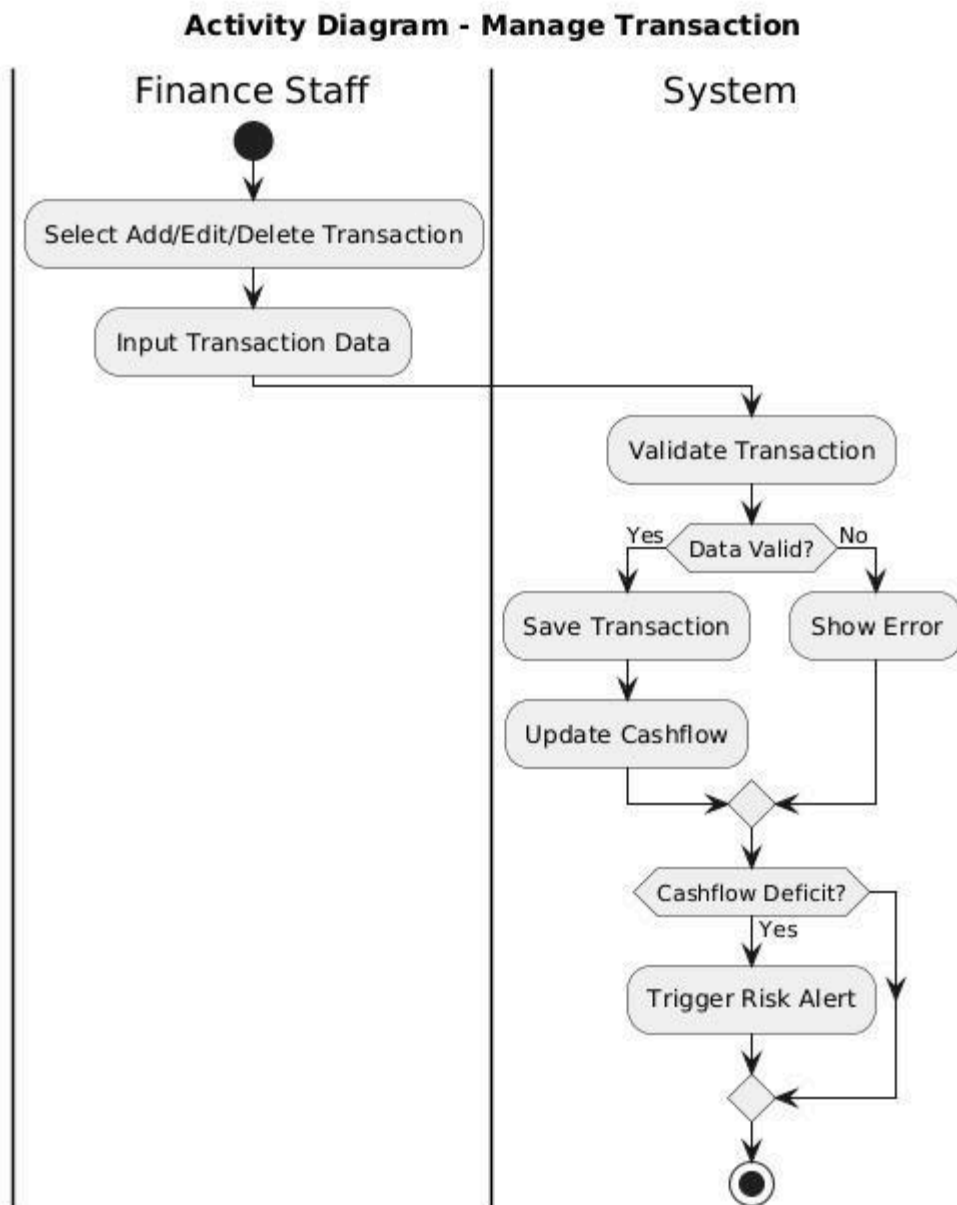
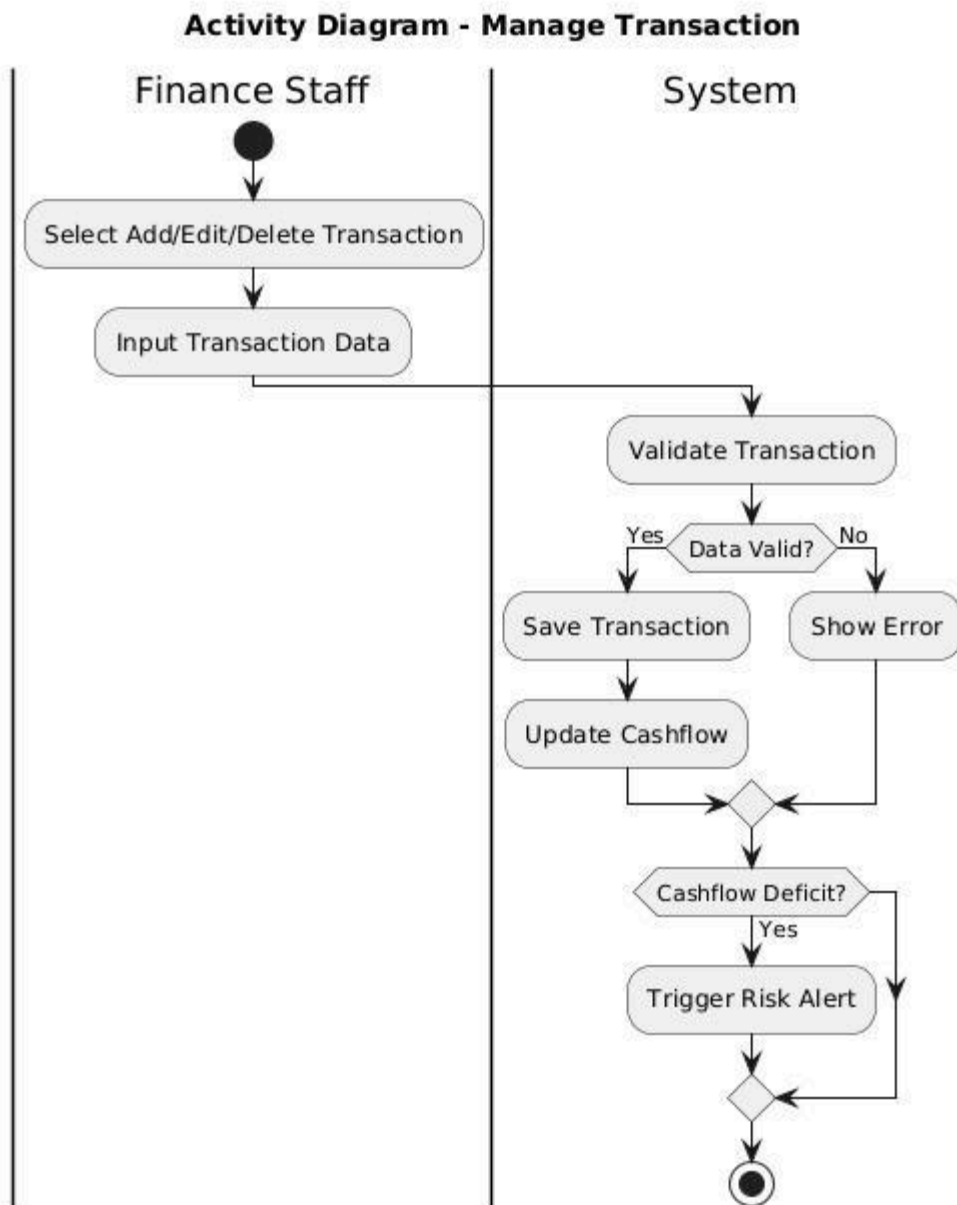
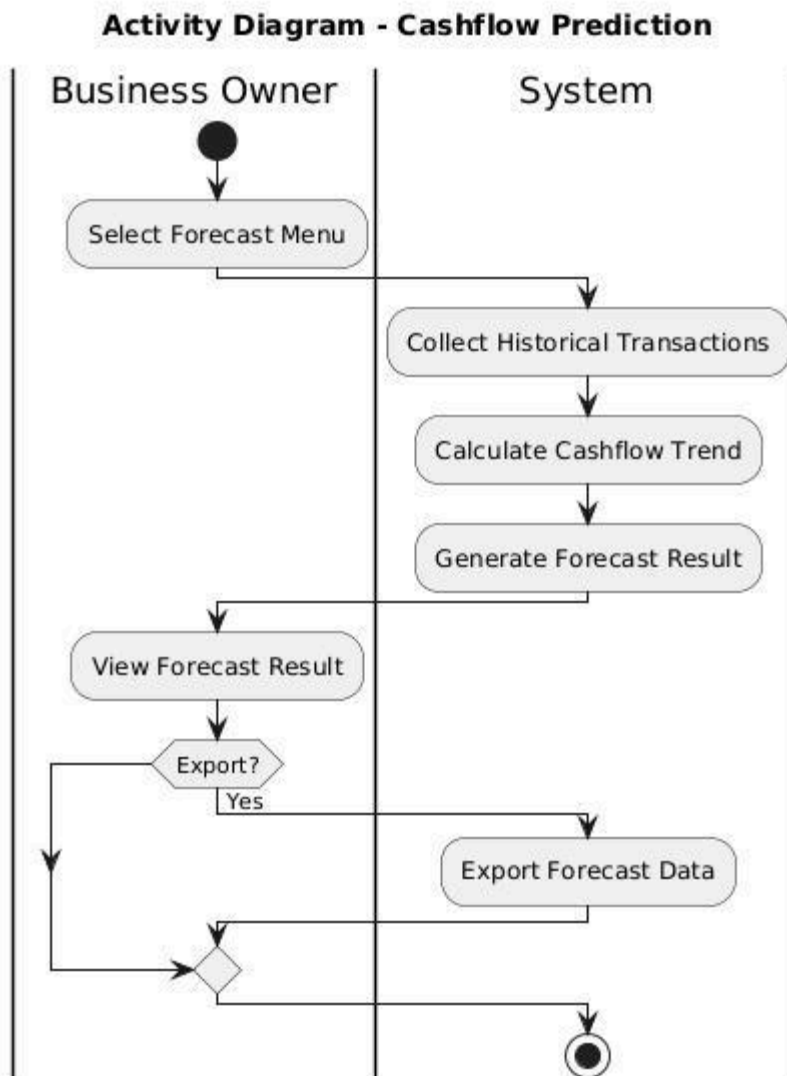


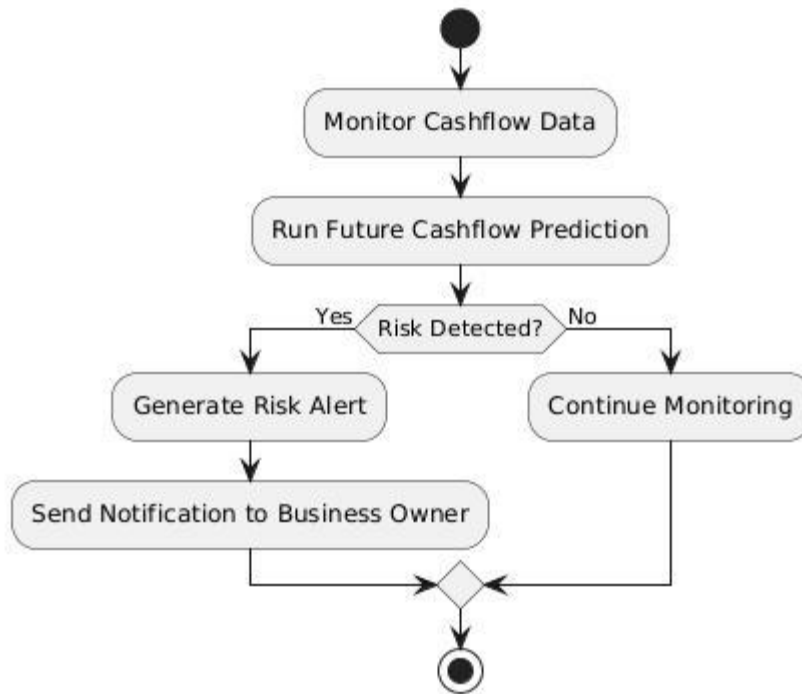
Figure 2.2 Activity Diagram - Manage Transaction



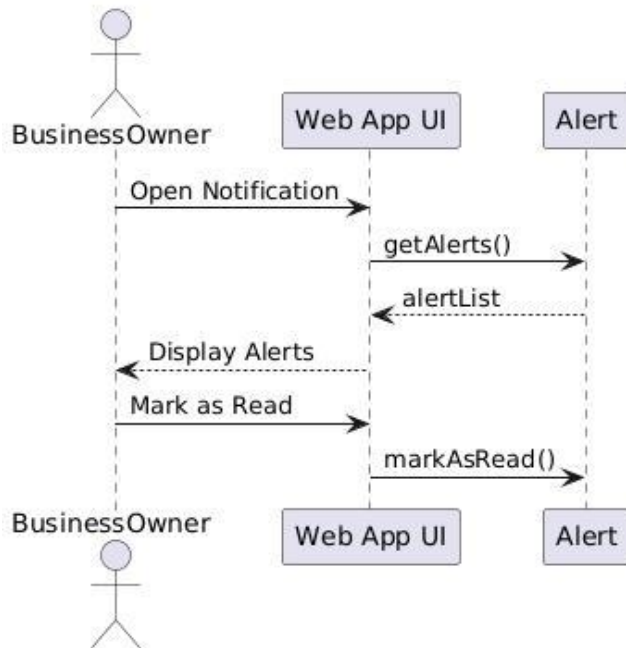
2.3 Activity Diagram -Manage Transaction



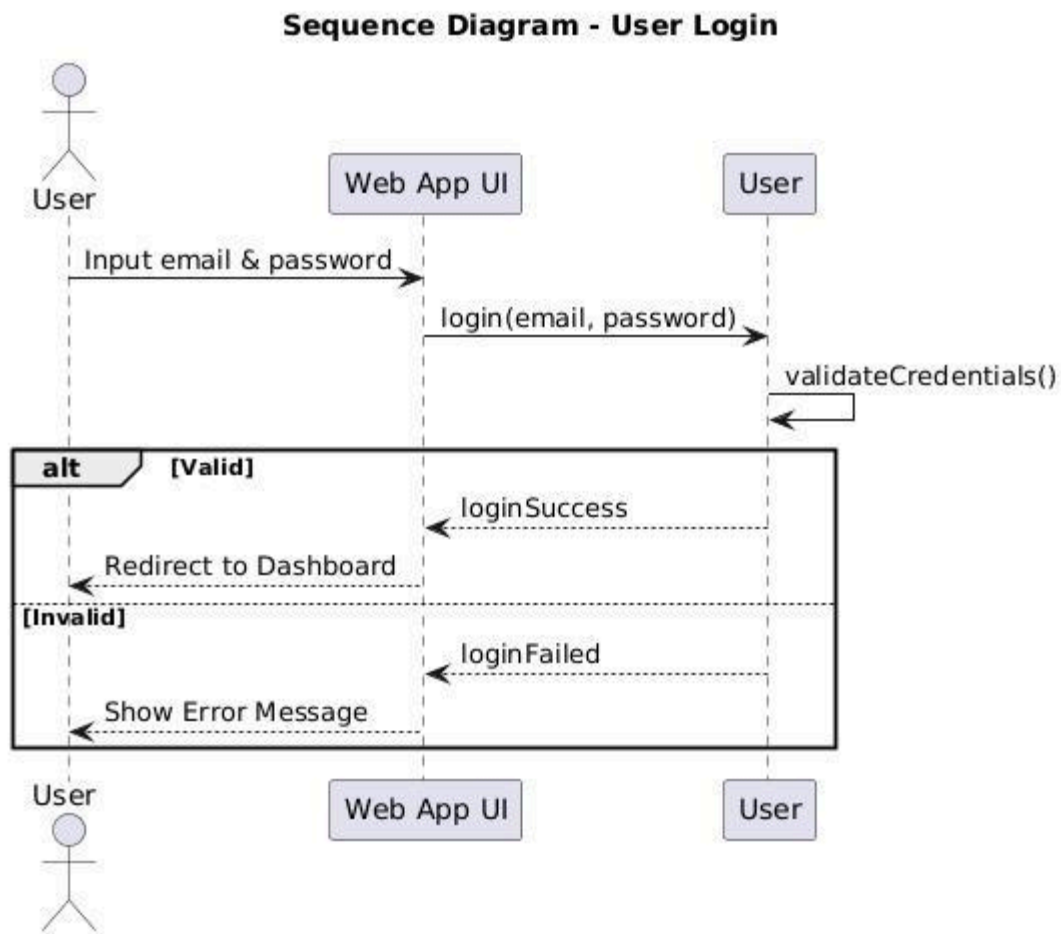
2.5 Activity Diagram - Cash Flow Prediction

Activity Diagram - Risk Notification System

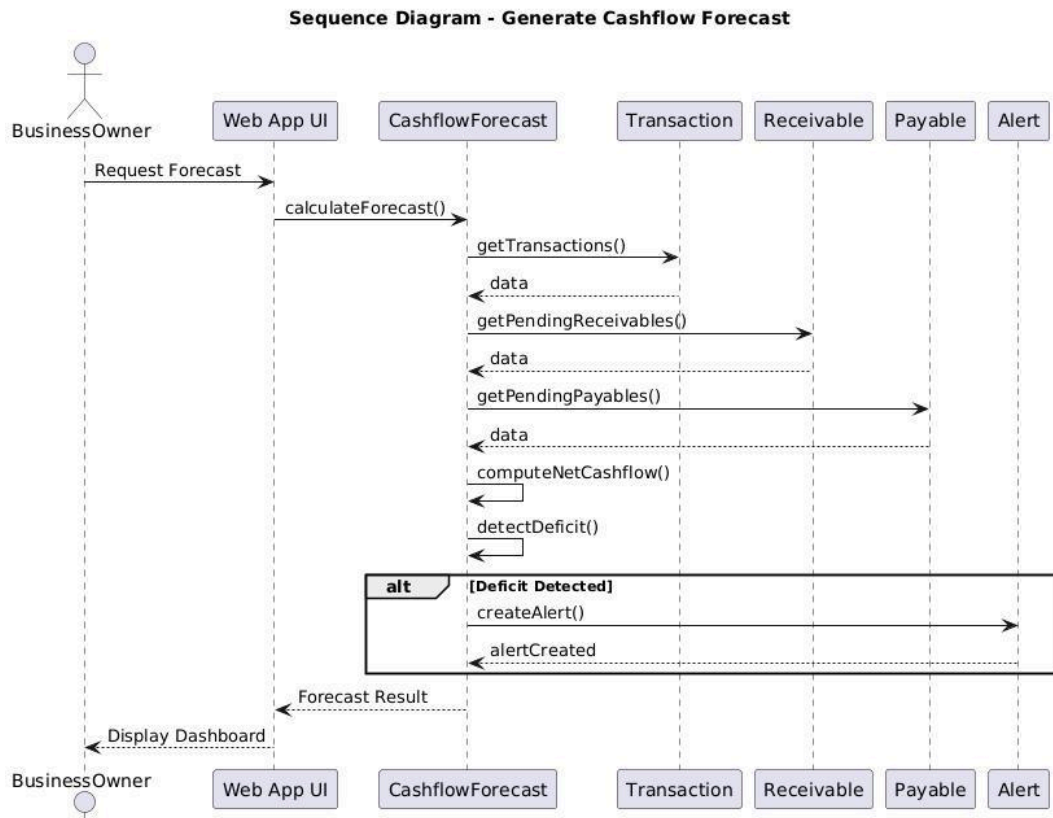
2.6 Activity Diagram - Risk Transaction

Sequence Diagram - View Alert

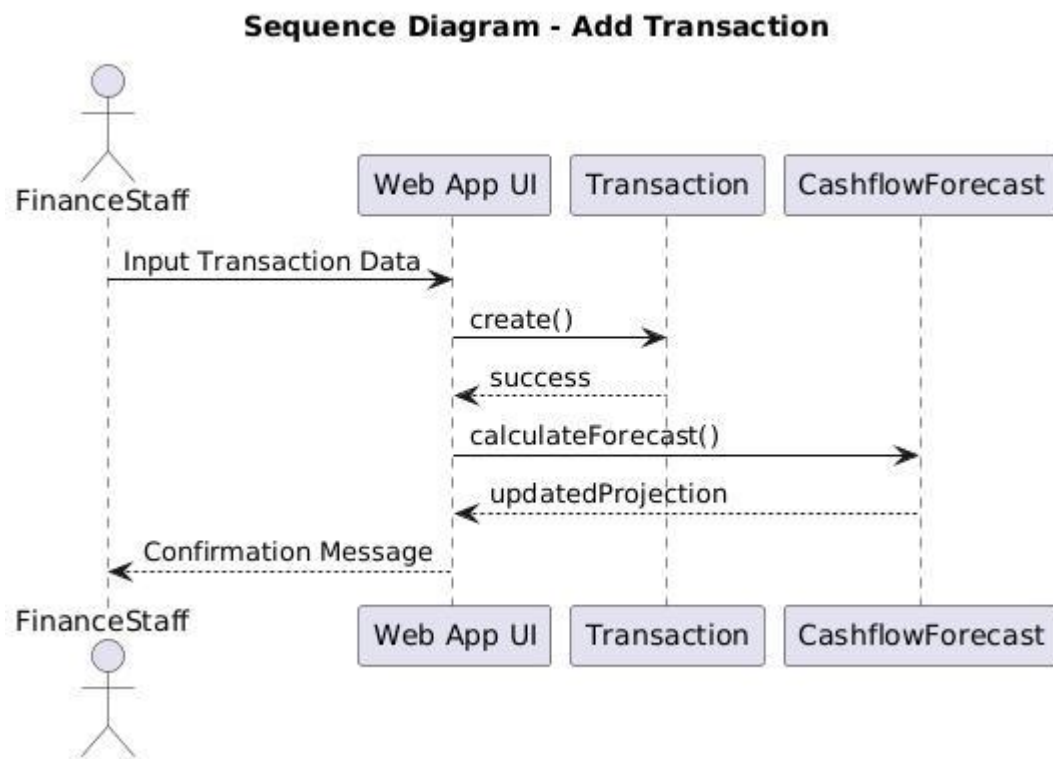
2.7 Sequence Diagram - View Alert



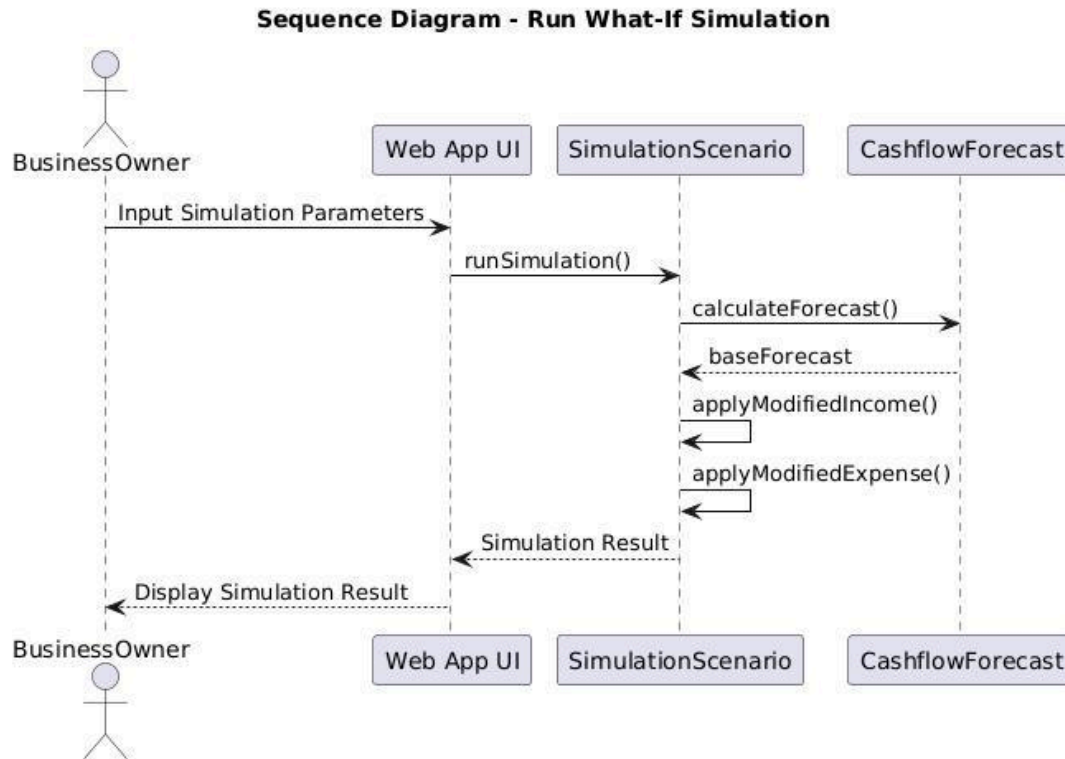
2.8 Sequence Diagram - User Login



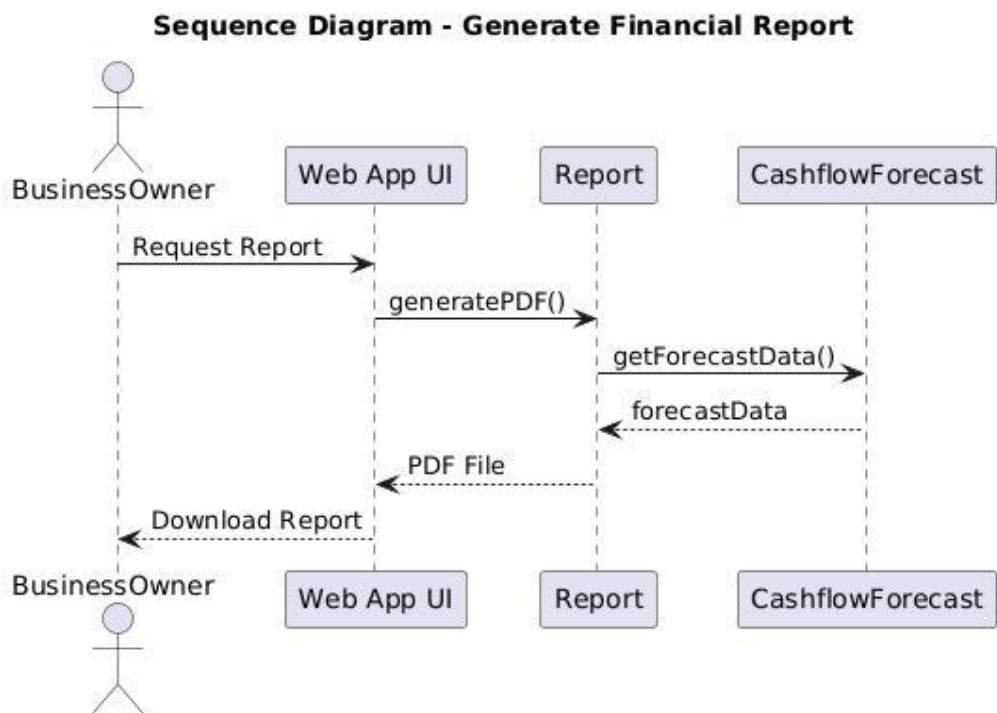
2.9 Sequence Diagram-Cashflow fore cast



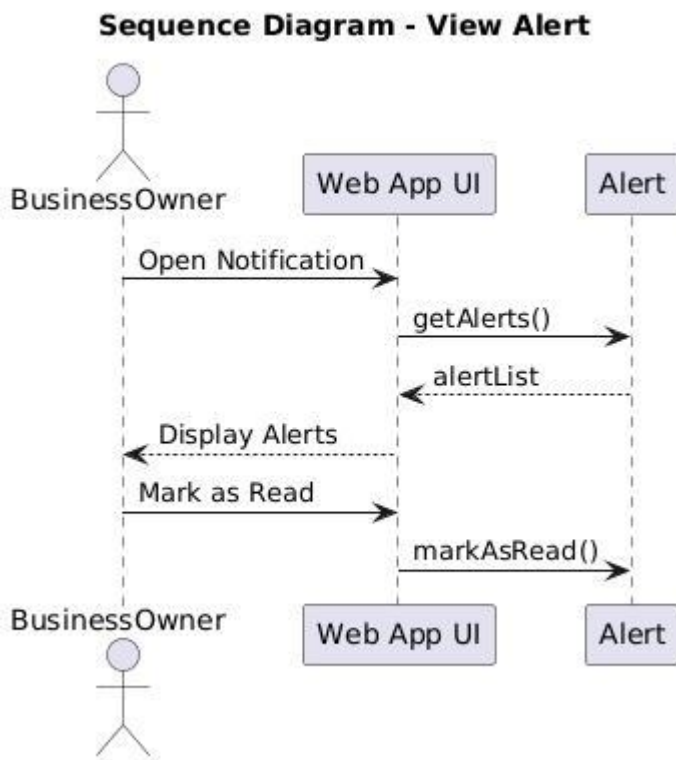
3.1 Sequence Diagram-Add Transaction



3.2 Sequence Diagram-What if

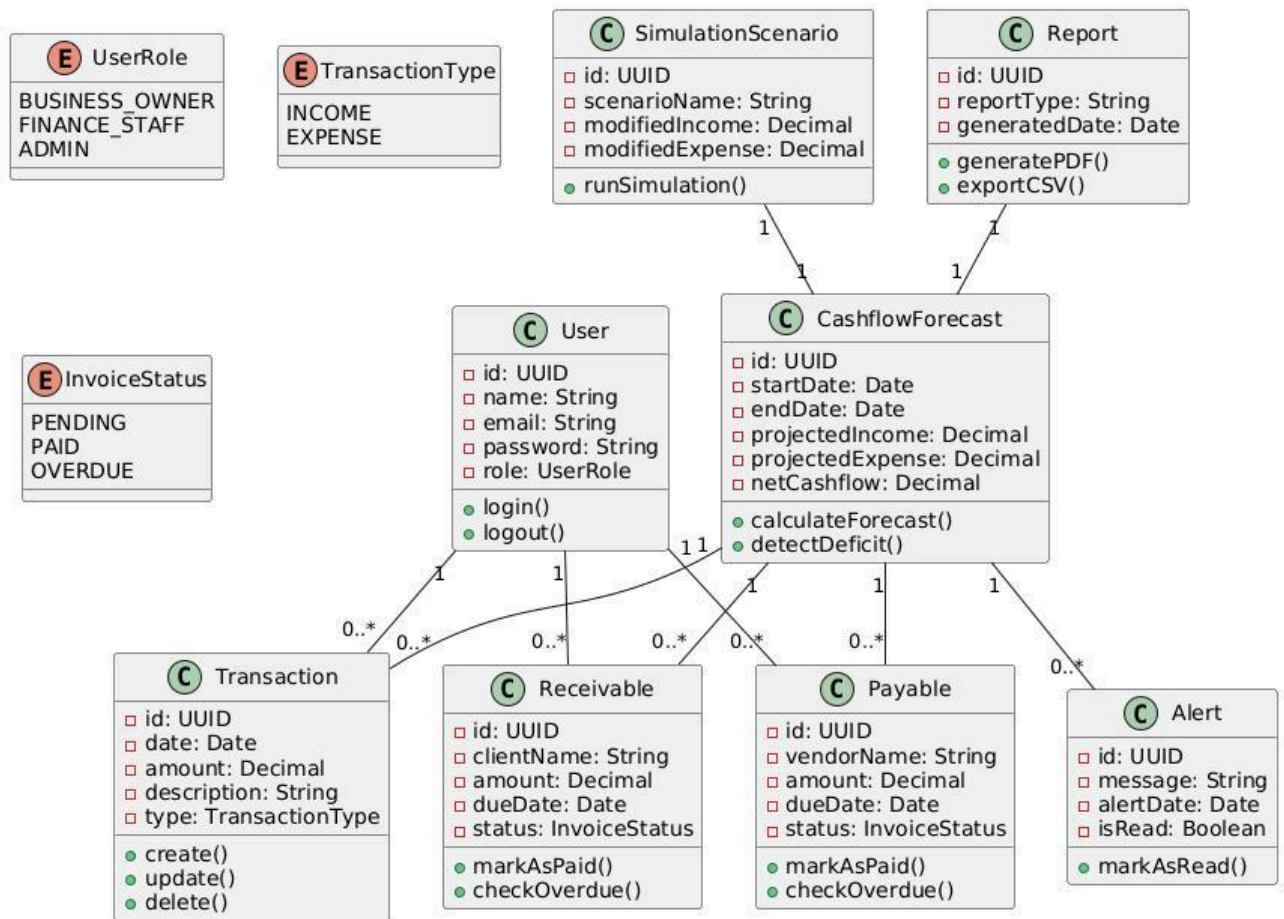


3.3 Sequence Diagram-Generate Financial Report



3.4 Sequence Diagram-View Alert

Business Cashflow Predictor System - Class Diagram



3.5 .Class Diagram-

2.3 User Classes and Characteristics

- **Business Owner:** Users with full access rights (Administrator) who need visibility into the overall cash flow to make strategic decisions. Technical expertise varies, so the interface must be highly intuitive.
- **Finance Staff / Operational Admin:** Users tasked with inputting daily transaction data (income/expenses). Their main focus is speed and accuracy in data entry.

2.4 Operating Environment

The system is a web-based application that will operate in the following environment:

- **Client:** Modern web browsers (Google Chrome, Mozilla Firefox, Safari, Microsoft Edge) on both desktop and mobile devices.
- **Server:** Cloud-based server with relational database support for transaction data storage.

2.5 Design and Implementation Constraints

- *initial phase (MVP) development is restricted to the core features of cash flow prediction and basic transaction management without direct third-party API integration.*
- *The system must be designed for easy future scalability to integrate complex analytics and open banking APIs*

2.6 User Documentation

FlowGuard akan dilengkapi dengan dokumentasi pengguna yang komprehensif untuk membantu pemilik bisnis dan staf dalam mengelola arus kas secara efektif. Dokumentasi ini akan disediakan dalam format digital yang dapat diakses langsung melalui platform.

User Manuals & Guides:

- **Business Owner's Guide:** Panduan mendalam untuk administrator yang mencakup pengaturan awal sistem, konfigurasi ambang batas peringatan defisit kas (*cash buffer*), dan cara menginterpretasikan analitik prediksi jangka panjang.
- **Staff's Guide:** Panduan operasional untuk tim keuangan yang menjelaskan langkah-langkah teknis seperti pencatatan transaksi harian, pengelolaan status hutang-piutang, dan pembaruan data inventaris.

Digital Help & Security Resources:

- **Online Help & Tutorials:** Panduan interaktif di dalam aplikasi (in-app guidance) mengenai cara melakukan sinkronisasi data antar perangkat dan cara menggunakan fitur simulasi skenario (*what-if analysis*).
- **FAQ Section:** Bagian tanya jawab yang menangani kendala teknis umum, visibilitas proyek, dan pengaturan izin akses tugas bagi pengguna dengan peran yang berbeda.
- **Kebijakan Privasi (Privacy Policy):** Dokumen legal dan teknis yang menjelaskan secara transparan bagaimana data keuangan bisnis dienkripsi di sisi klien sebelum diunggah ke *cloud* untuk menjamin keamanan data sensitif perusahaan.

Project Delivery Components: Selain dokumentasi pengguna di atas, proyek ini juga akan menyertakan dokumen teknis berikut:

- **Software Requirements Specification (SRS):** Dokumen ini, yang merinci seluruh kebutuhan fungsional dan non-fungsional.
- **User Interface Specification:** Dokumen terpisah yang berisi desain UI mendetail dan tata letak layar bertema *nebula*.
- **Prototype Simulation Guide:** Panduan yang menjelaskan fitur mana yang sudah berfungsi penuh dan fitur mana yang masih berupa simulasi dalam lingkungan prototipe.

2.7 Assumptions and Dependencies

It is assumed that users (SMEs) have a stable internet connection to access the application in real-time.

The quality and accuracy of cash flow predictions heavily depend on the discipline of the user's staff in routinely entering historical transaction data.

3. External Interface Requirements

3.1 User Interfaces

The user interface will be designed using a nebula-themed color palette, providing a modern and elegant visual impression. The system will implement a responsive design so that the analytics dashboard layout and input forms adapt seamlessly to both desktop and mobile screens. All pages will feature a persistent navigation bar to access the Dashboard, Transactions, and Predictions. Error messages or input validations will be displayed in contrasting colors to alert users immediately.

3.2 Hardware Interfaces

There is no interaction with specific custom hardware. The system operates over standard HTTP web client-server interactions.

3.3 Software Interfaces

- **Database Engine:** Used to store all user relational data, income/expense transactions, and prediction logs.
- **Future External APIs:** For advanced development phases, the system will prepare endpoints capable of interacting with banking APIs or third-party accounting software.

3.4 Communications Interfaces

All communication between the web client and server must use the HTTPS protocol to ensure the encryption of business financial data transmission.

4. System Features

<This template illustrates organizing the functional requirements for the product by system features, the major services provided by the product. You may prefer to organize this section by use case, mode of operation, user class, object class, functional hierarchy, or combinations of these, whatever makes the most logical sense for your product.>

4.1 System Feature 1

4.1.1 Description and Priority

A core feature (High Priority) to project cash availability for the next 30–90 days

4.1.2 Stimulus/Response Sequences

The user navigates to the "Predictions" page. The system processes historical data and scheduled transactions, then displays the cash flow trend chart.

4.1.3 Functional Requirements

REQ-1: The system must be able to calculate daily/weekly cash flow predictions for up to 90 days ahead.

REQ-2: The system must use past transaction data and scheduled accounts payable/receivable as calculation variables.

4.2 Income and Expense Management Module

4.2.1 Description and Priority

A fundamental feature (High Priority) for structured transaction recording.

4.2.2 Stimulus/Response Sequences

The user clicks the "Add Transaction" button, fills out the form, and saves it. The system updates the balance and database.

4.2.3 Functional Requirements

REQ-3: The system must provide input forms for income, expenses, accounts receivable, and accounts payable.

REQ-4: The system must differentiate transaction statuses (paid, overdue, scheduled).

4.3 Notification And Alert System

4.3.1 an early warning feature (Medium-High Priority) to prevent operational deficits.

4.3.2 Stimulus/Response Sequences

f the prediction results detect that the cash balance will drop below a safe threshold, the system will trigger an alert on the user's dashboard.

4.3.3 Functional Requirements

REQ-5: The system must display a visual early warning if the projected cash balance hits a negative figure during a specific period.

4.4 Financial Scenario Simulation Module

4.4.1 An analytical feature (Medium Priority) for testing financial decisions (what-if analysis).

4.4.2 **REQ-6:** The system must allow users to input hypothetical variables (e.g., "what if production equipment A is purchased next month?") without altering actual data.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system must load the main dashboard along with analytical chart calculations within a maximum of 3 seconds under standard network conditions. Complex prediction calculations (90 days ahead) must not block the user interface and should preferably be executed asynchronously in the background.

5.2 Safety Requirements

In the event of a server failure, the system must have an automatic database recovery mechanism to ensure no financial records are corrupted, lost, or duplicated (data integrity principle).

5.3 Security Requirements

- All user authentication must be encrypted using industry-standard algorithms (e.g., Bcrypt).
- The system must adhere to tenant isolation practices so that individual business data cannot be accessed without authorization by accounts from other business entities.

5.4 Software Quality Attributes

- **Usability:** Highly crucial as this application aims to minimize error rates compared to manual spreadsheet tracking. Navigation must be intuitive for non-accountants.
- **Reliability:** Service availability (Uptime) must be high to reduce the risk of business operational disruptions when monitoring payment obligations.

5.5 Business Rules

Only users with the "Business Owner/Administrator" role are permitted to modify initial configuration settings (such as minimum buffer thresholds for the cash deficit warning system). Operational staff are restricted solely to transaction data input.

6. Other Requirements

<Define any other requirements not covered elsewhere in the SRS. This might include database requirements, internationalization requirements, legal requirements, reuse objectives for the project, and so on. Add any new sections that are pertinent to the project.>

Appendix A: Glossary

<Define all the terms necessary to properly interpret the SRS, including acronyms and abbreviations. You may wish to build a separate glossary that spans multiple projects or the entire organization, and just include terms specific to a single project in each SRS.>

Appendix B: Analysis Models

<Optionally, include any pertinent analysis models, such as data flow diagrams, class diagrams, state-transition diagrams, or entity-relationship diagrams.>

Appendix C: To Be Determined List

<Collect a numbered list of the TBD (to be determined) references that remain in the SRS so they can be tracked to closure.>